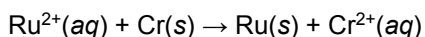
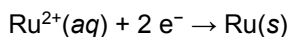


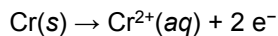
In a proposed theoretical galvanic cell, the overall redox reaction:



can be expressed as separate reduction and oxidation processes (Half-reactions 1 and 2).



Half-reaction 1: $E^{\circ}_{\text{red}} = +0.46 \text{ V}$



Half-reaction 2: $E^{\circ}_{\text{ox}} = +0.90 \text{ V}$

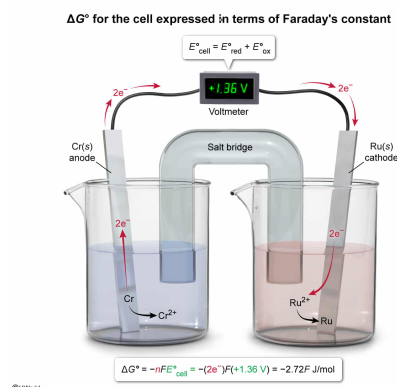
What is the expected change in the standard Gibbs free energy ΔG° for the galvanic cell expressed in terms of Faraday's constant F ?

- ☐ A. $-0.92F \text{ J/mol}$ [10%]
☐ B. $-1.36F \text{ J/mol}$ [44%]
☐ C. $-1.80F \text{ J/mol}$ [9%]
☒ D. $-2.72F \text{ J/mol}$ [35%]

Correct

34% answered correctly

Explanation:

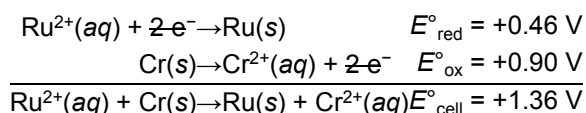


In an electrochemical cell, an **oxidation-reduction (redox) reaction** converts chemical energy into electrical energy (or vice versa). The change in the standard **Gibbs free energy** ΔG° associated with the redox reaction is proportional to the standard cell potential E°_{cell} according to the equation:

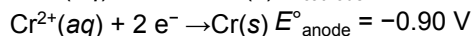
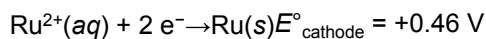
$$\Delta G^{\circ} = -nFE^{\circ}_{\text{cell}}$$

where n is the number of electrons transferred per reaction and F is **Faraday's constant**. As such, a **positive** E°_{cell} indicates a thermodynamically **favorable reaction** suitable for a **galvanic cell** whereas a **negative** E°_{cell} indicates an **unfavorable reaction** that can only be achieved by an **electrolytic cell** in which external energy is supplied to the system.

For the given galvanic cell, the net reaction is found by adding the reduction and oxidation half-reactions in such a way that the transferred electrons cancel.



In many data tables, the convention is to write all half-reactions as *reductions* with the electrons on the reactants side of the half-reaction. If this were done for this system, the half-reactions would be shown with their corresponding *reduction* potentials as:



In such cases, E_{cell}° is found by:

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

$$E_{\text{cell}}^{\circ} = 0.46 \text{ V} - (-0.90 \text{ V}) = 0.46 \text{ V} + 0.90 \text{ V} = +1.36 \text{ V}$$

This equation assumes that both values are *reduction* potentials. If one of the half-reactions is instead written as an *oxidation* with the electrons on the products side, the half-reaction is flipped and the sign of its potential will be the *opposite* to that of the reduction (eg, positive instead of negative). Because one of the given half-reactions in this question is written as an *oxidation* (with its sign already switched) and the other is written as a reduction, the value of E_{cell}° is evaluated as the sum of E_{red}° and E_{ox}° .

$$E_{\text{cell}}^{\circ} = E_{\text{red}}^{\circ} + E_{\text{ox}}^{\circ}$$

$$E_{\text{cell}}^{\circ} = 0.46 \text{ V} + 0.90 \text{ V} = +1.36 \text{ V}$$

Therefore, ΔG° for the galvanic cell (expressed in terms of [Faraday's constant](#)) is:

$$\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ}$$

$$\Delta G^{\circ} = -(2 \text{ e}^{-}) \left(F \frac{\text{J}}{\text{V} \cdot \text{mol e}^{-}} \right) (+1.36 \text{ V})$$

$$\Delta G^{\circ} = -(2)(+1.36) \left(F \frac{\text{J}}{\text{mol}} \right) = -2.72F \frac{\text{J}}{\text{mol}}$$

Finishing the calculation to a numerical result (although not required) gives:

$$\Delta G^{\circ} = -2.72 \left(96,485 \frac{\text{J}}{\text{mol}} \right) = 262,439 \frac{\text{J}}{\text{mol}} \Rightarrow 262 \frac{\text{kJ}}{\text{mol}}$$

(Choice A) A value of $-0.92F \text{ J/mol}$ is obtained if E_{red}° is used in the calculation instead of E_{cell}° .

(Choice B) A value of $-1.36F \text{ J/mol}$ is obtained if a transfer of one electron ($n = 1$) is used in the calculation. Half-reactions 1 and 2 indicate that a transfer of *two electrons* ($n = 2$) occurs in the net reaction within the cell.

(Choice C) A value of $-1.80F \text{ J/mol}$ is obtained if E_{ox}° is used in the calculation instead of E_{cell}° .

Educational objective:

The Gibbs free energy ΔG° of an electrochemical cell is related to the number of electrons n transferred in the overall reaction, the overall cell potential E_{cell}° , and Faraday's constant F , according to the equation $\Delta G^{\circ} = -nFE_{\text{cell}}^{\circ}$. A positive E_{cell}° indicates a thermodynamically favorable reaction whereas a negative E_{cell}° indicates an unfavorable reaction.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403890

System:
Solutions and Electrochemistry